

What is "synthetic spectra"?

$$\mathrm{Br}(E_n) \leftarrow$$

Three models: 1. spherical sheaves.

$$\mathcal{P}_\Sigma(\mathrm{Mod}_E)$$

$$\downarrow \text{preserves finite limits.}$$

$$\mathcal{P}(\mathrm{Mod}_E)$$

2. filtered spectra.

$$\mathrm{Sp} \rightarrow \mathrm{Fil}(\mathrm{Sp}).$$

3. motivic cellular modules.

k field k -motivic stable homotopy category

$$\mathrm{SH}(k) = \left[\mathrm{Shv}(\mathrm{Sm}_k) \right]_{\mathrm{mot}} \cdot [\mathbb{P}_k^{1-1}].$$

$\mathbb{Z}$ poset, $N(\mathbb{Z})$

$$\mathrm{Fun}(N(\mathbb{Z}), \mathrm{Sp}) =: \mathrm{Fil}(\mathrm{Sp}) \leftarrow \mathrm{HA}$$

$$\mathrm{Fun}(N(\mathbb{Z}^{\mathrm{op}}), \mathrm{Sp}) =: \mathrm{Fil}(\mathrm{Sp})$$

In general, Sp may be replaced by any stable ∞ -cat $\mathcal{C}$ with t -structure.

$$\cdots \rightarrow X_n \rightarrow X_{n-1} \rightarrow X_{n-2} \rightarrow \cdots$$

$\mathcal{C}^{\mathbb{B}}$ abelian

HA tell us that we have a spectral seq.

cofiber
 $\downarrow$

$$\pi_{p+q} \varinjlim_n X_n \leftarrow E_1^{p,q} \simeq \pi_{p+q} \left(\frac{X_p}{X_{p-1}} \right)$$

What is "Segal conjecture"?

Def A cyclotomic spectrum is a pair of $\mathbb{T}$ -spectrum X and family $(Y_p)_{p \in \mathbb{P}}$.

$$\varphi_p: X \rightarrow X^{+G} \text{ is } \mathbb{T} \text{-equivariant.}$$

$$\varphi_p: X \longrightarrow X^{+p} \quad \text{is } H\text{-equivariant.}$$

$$X_{hC_p} \xrightarrow{Nm} X^{hC_p} \longrightarrow X^{+C_p} \quad X = HM$$

group homology group cohomology

Segal conjecture has two forms.

Setting: $X \in C_{yc} S_p$, X bounded below, fix a prime p .

Form 1

$$\begin{array}{ccc} \varphi_p: X & \longrightarrow & X^{+C_p} \\ \downarrow & \nearrow & \uparrow \\ X_p^1 & & \varphi_p' \end{array} \quad \begin{array}{l} \text{NS19} \\ (p\text{-complete}) \end{array}$$

φ_p' is an equivalence.

Ex. $S^{\text{triv}} \simeq THH(S)$, $\text{Lin } (p=2)$.

$THH(S)$ admits Segal conjecture, for any p .

$$\varphi_p: S \longrightarrow S^{+C_p} \quad \text{is } p\text{-adic equivalence.}$$

$\varphi_p \otimes \frac{S}{p}$ is equivalence

Form 2

Chromatic K -theory.

Setting: $\left[\begin{array}{l} \text{Fix a prime } p, \text{ type } n \text{ finite complex } F, \\ X \in C_{yc} S_p \end{array} \right]$

F -Segal conjecture for X

$$\text{the map } F \otimes \varphi_p: F \otimes X \longrightarrow F \otimes X^{+C_p}$$

has bounded above fiber.

i.e. $\pi_* F \otimes \varphi_p$ is isomorphism for $* \gg 0$.

Main Result:

Thm (HW 22) Let F denote any type $n+1$ finite complex,

then $\boxed{F\text{-Segal conjecture for } THH(B_p\langle n \rangle)}$

holds. $THH_*(B_p\langle n \rangle) \xrightarrow[\substack{(p, v_1 \dots v_n) \\ \text{is iso for } * \gg 0}}]{\varphi'_p} THH_*(B_p\langle n \rangle)^{tG_p}$

Remark:

① For any $R \in \text{Alg}(S_p)$

$$THH(R) := \left| \begin{array}{c} \cdots R^{\otimes 3} \rightrightarrows R^{\otimes 2} \rightrightarrows R \\ \vdots \end{array} \right| \in \text{Alg}(e) \left\{ \begin{array}{c} \text{Colim}_{m \rightarrow \infty} R^{\otimes m} \\ \text{Alg}(e) \end{array} \right\} \in \text{Alg}(e)$$

Fact: $THH(R) \in \text{Cyc } S_p$.
p-local

② Let $n \geq 0$, $\exists \mathbb{E}_3\text{-MU-alg } B$ s.t.

$$\mathbb{Z}_{(p)}[v_1 \dots v_n] \subseteq B_{p*} \subseteq (MU_{(p)})_* \rightarrow \pi_* B.$$

is an isomorphism.

So, the symbol " $B_p\langle n \rangle$ " is the $\mathbb{E}_3\text{-MU-alg } B$.

Strategy:

• Any type $n+1$ finite complex is thick sub cat generated by $\frac{\$}{(p^{i_0}, v_1^{i_1} \dots, v_n^{i_n})}$.

• $\frac{\$}{(p^{i_0}, v_1^{i_1} \dots, v_n^{i_n})} \otimes THH(B_p\langle n \rangle) \rightarrow \frac{\$}{(p^{i_0}, v_1^{i_1} \dots, v_n^{i_n})} \otimes THH(B_p\langle n \rangle)^{tG_p}$

$$\frac{+}{(p_1^{i_0}, v_1^{i_1}, \dots, v_n^{i_n})} \otimes HH(Bp\langle n \rangle) \rightarrow \frac{-}{(p_1^{i_0}, v_1^{i_1}, \dots, v_n^{i_n})} \otimes HH(Bp\langle n \rangle)'$$

$$\frac{Bp\langle n \rangle}{(p, v_1, \dots, v_n)} \simeq \mathbb{F}_p.$$

It suffices to show that

$$\mathbb{F}_p \otimes_{Bp\langle n \rangle} THH(Bp\langle n \rangle) \rightarrow \mathbb{F}_p \otimes_{Bp\langle n \rangle} THH(Bp\langle n \rangle)^{t\mathbb{G}_p}$$

has bounded above fibres. $\langle \langle \mathbb{Q}$ -motivic modular forms $\rangle \rangle$

$$\text{desc}_{\mathbb{F}_p} : Sp \rightarrow \text{Fil}(Sp) \quad \text{last symmetric monoidal}$$

$$\lim_{\Delta} (C_{\geq} (Bp\langle n \rangle \otimes_{\mathbb{F}_p} \mathbb{F}_p^{\otimes [n]})) \quad \text{Adams}_{\mathbb{F}_p} Bp\langle n \rangle \simeq \lim_{i \in \Delta, m} Bp\langle n \rangle \otimes_{\mathbb{F}_p} \mathbb{F}_p^{\otimes [m]}$$

$$\text{desc}_{\mathbb{F}_p} \mathbb{F}_p \otimes_{\text{desc}_{\mathbb{F}_p} Bp\langle n \rangle} THH(\text{desc}_{\mathbb{F}_p} Bp\langle n \rangle) \rightarrow \text{desc}_{\mathbb{F}_p} \mathbb{F}_p \otimes_{\text{desc}_{\mathbb{F}_p} Bp\langle n \rangle} THH(-)^{t\mathbb{G}_p}$$

$$\text{desc}_{\mathbb{F}_p} \mathbb{B} \quad \text{sit} \quad E_2 \quad \text{is} \quad \mathbb{F}_p \otimes_{Bp\langle n \rangle} THH(Bp\langle n \rangle) \xrightarrow{\wedge} \mathbb{F}_p \otimes_{Bp\langle n \rangle} THH(Bp\langle n \rangle)^{t\mathbb{G}_p}$$

$$\text{Ext}_{A_{+,*}}(\mathbb{F}_p, H_*(X; \mathbb{F}_p)) \quad \text{is} \quad \mathbb{F}_p \otimes_{Bp\langle n \rangle} THH(Bp\langle n \rangle) \rightarrow \mathbb{F}_p \otimes_{Bp\langle n \rangle} THH(Bp\langle n \rangle)^{t\mathbb{G}_p}$$

$$\pi_{+,-s}(\text{gr}^{t\text{desc}_{\mathbb{F}_p}} X) \quad \pi_* Bp\langle n \rangle \otimes_{\mathbb{F}_p} \simeq A \square_{\wedge(\bar{v}_1, \dots, \bar{v}_n) \mathbb{F}_p} \mathbb{F}_p$$

$$\pi_* (\text{gr}^{\text{desc}_{\mathbb{F}_p}} Bp\langle n \rangle) \simeq \text{Ext}_A(\mathbb{F}_p, A \square_{\wedge(\bar{v}_1, \dots, \bar{v}_n) \mathbb{F}_p} \mathbb{F}_p)$$

$$\text{change} \quad \text{Ext}_{\wedge(\bar{v}_1, \dots, \bar{v}_n)}^{\text{sit}}(\mathbb{F}_p, \mathbb{F}_p).$$

$$\simeq \mathbb{F}_p[v_1, \dots, v_n].$$

We denote $v_i = 2(p^{i-1})$. $\text{gr } Sp$

We denote $\text{gr } B_p(n) \rightarrow R \in \text{Alg}_{\mathbb{E}_2}(\text{gr } Sp)$ $\text{gr } Sp := \text{Fun}(\mathbb{Z}^{de}, Sp)$

Prop: R is equiv to $\mathbb{F}_p \otimes \mathcal{S}[a_1] \otimes \dots \otimes \mathcal{S}[a_n]$, as $\mathbb{E}_2$ - $\mathbb{F}_p$ -graded alg.

where $\mathcal{S}[a_i]$ is the free graded $\mathbb{E}_2$ -alg generated by $\mathcal{S}^{2(p^i-1)}(w_i) = \begin{cases} 0, & j \neq w_i \\ \mathcal{S}^{2(p^i-1)}, & j = w_i \end{cases} \in \text{gr}(Sp)$

Sketch: Only need to construct a $\mathbb{E}_2$ -alg, $\exists \mathbb{E}_2$ $R \rightarrow A$

Let $A = \mathbb{F}_p \otimes \mathcal{S}[a_1] \otimes \dots \otimes \mathcal{S}[a_n]$. only need to prove $\mathbb{F}_p \otimes \mathcal{S}[a_1] \otimes \dots \otimes \mathcal{S}[a_n] \xrightarrow{\cong} \mathbb{F}_p \otimes \mathbb{F}_p \otimes \mathbb{F}_p \otimes \mathbb{F}_p$ is even $\left(\begin{smallmatrix} \text{Cellular} \\ \mathbb{E}_k\text{-alg} \end{smallmatrix} \right)$

$$\pi_* \mathbb{F}_p \otimes_{A \otimes \mathbb{F}_p} \mathbb{F}_p \simeq \bigwedge_{\mathbb{F}_p} (\sigma_{a_1}, \dots, \sigma_{a_n}) \quad (\text{Thm of } A_\infty\text{-mg Spectra})$$

$$\text{Tor}_{\mathbb{F}_p}^{\bigwedge_{\mathbb{F}_p} (\sigma_{a_1}, \dots, \sigma_{a_n})} (\mathbb{F}_p, \mathbb{F}_p) \Rightarrow \pi_* \mathbb{F}_p \otimes_{\mathbb{F}_p \otimes \mathbb{F}_p} \mathbb{F}_p \otimes_{\mathbb{F}_p \otimes \mathbb{F}_p} \mathbb{F}_p$$

$$\bigwedge_{\mathbb{F}_p} \langle \sigma^2 a_1, \dots, \sigma^2 a_n \rangle \text{ van } \quad \square$$

$$\text{des}_{\mathbb{F}_p} B_p(n) = \lim_{[i] \in \Delta} (\varprojlim_{j \geq i} B_p(n) \otimes \mathbb{F}_p^{\otimes [i]})$$

$$\text{Adm}_{\mathbb{F}_p} B_p(n) = \lim_{[i] \in \Delta_{\leq}} (B_p(n) \otimes \mathbb{F}_p^{\otimes [i]})$$

$\text{Fil}(Sp)$ has a $+$ -structure, Beilinson t -structure, $\tau_{\geq \bullet}^B$ $\tau_{\leq \bullet}^B$ $\text{Adm}_{\mathbb{F}_p} B_p(n)$ HW 22.

$$\tau \geq \cdot$$

$$\tau \geq * \text{Adm}_{\mathbb{F}_p} B_p(n)$$

HW 22 -
C-motivic
Anisotropy

$$\text{Prop} \quad \frac{\text{THH}_* (\mathbb{F}_p[v_1 \dots v_n])}{(v_1, \dots, v_n)} \rightarrow \frac{\text{THH}_*^{\pm \mathbb{C}_p} (\mathbb{F}_p[v_1 \dots v_n])}{(v_1, \dots, v_n)}$$

is an equiv. in degrees $* > \sum_{i=1}^n |\mathbb{N}_i| n$

Sketch:

$$\text{Fact: } 1) \text{THH} = \text{Alg}(\text{gr } S_p) \rightarrow \text{gr } S_p$$

is symmetric monoidal.

$$2) L_p \text{THH}(\mathbb{S}[a_i]) \rightarrow \text{THH}(\mathbb{S}[a_i])^{\pm \mathbb{C}_p}$$

witness the right hand side as p-completion of left hand side.

$$\text{THH}(\mathbb{S}[a_i]) \simeq \bigoplus_{m \geq 0} \mathbb{S}^{|a_i|}(w) \wedge \mathbb{T}/C_m$$

$$\text{THH}(\mathbb{F}_p \otimes \mathbb{S}[a_1] \otimes \dots \otimes \mathbb{S}[a_n])$$

$$\downarrow f_1 \simeq \text{by 1) (symmetric monoidal)}$$

$$\text{THH}(\mathbb{F}_p) \otimes \text{THH}(\mathbb{S}[a_1]) \otimes \dots \otimes \text{THH}(\mathbb{S}[a_n])$$

$$\downarrow f_2$$

$$\text{THH}(\mathbb{F}_p)^{\pm \mathbb{C}_p} \otimes \text{THH}(\mathbb{S}[a_1])^{\pm \mathbb{C}_p} \otimes \dots \otimes \text{THH}(\mathbb{S}[a_n])^{\pm \mathbb{C}_p}$$

$$\downarrow f_3$$

$$\left(\text{THH}(\mathbb{F}_p) \otimes \dots \otimes \text{THH}(\mathbb{S}[a_n]) \right)^{\pm \mathbb{C}_p} \simeq \text{THH}(\mathbb{F}_p[v_1 \dots v_n])^{\pm \mathbb{C}_p}$$

3) f_3 is equivalence.

It suffices to show modulo $(a_1, \dots, a_n)$ f_* is isomorphism.

Note that
$$T\mathcal{H}(\mathbb{F}_p)^{+C}_p \otimes_{\mathbb{F}_p} \mathbb{F}_p \otimes \bigotimes_{i=1}^n T\mathcal{H}(S[a_i])^{+C}_p$$

$$\cong T\mathcal{H}(\mathbb{F}_p)^{+C}_p \otimes_{\mathbb{F}_p} \mathbb{F}_p \otimes \bigotimes_{i=1}^n T\mathcal{H}(S[a_i])^{+C}_p$$

$$\cong T\mathcal{H}(\mathbb{F}_p)^{+C}_p \otimes_{\mathbb{F}_p} \mathbb{F}_p \bigotimes_{i=1}^n T\mathcal{H}(S[a_i])$$

Note that $\forall X \in S_p, X_p^* \otimes_{\mathbb{F}_p} \mathbb{F}_p \cong X \otimes_{\mathbb{F}_p} \mathbb{F}_p$

By $\mathbb{F}_p \oplus \mathbb{Z} \cong \mathbb{F}_p \Rightarrow$

$$T\mathcal{H}(\mathbb{F}_p) \otimes_{\mathbb{F}_p} \mathbb{F}_p \otimes \bigotimes_{i=1}^n T\mathcal{H}(S[a_i]) \rightarrow T\mathcal{H}(\mathbb{F}_p)^{+C}_p \otimes_{\mathbb{F}_p} \mathbb{F}_p \otimes \bigotimes_{i=1}^n T\mathcal{H}(S[a_i])$$

$$\mathbb{F}_p[u] \otimes_{\mathbb{F}_p} \bigotimes_{i=1}^n \mathcal{H}\mathcal{H}\left(\frac{\mathbb{F}_p[a_i]}{\mathbb{F}_p}\right) \rightarrow \mathbb{F}_p[u^{\pm 1}] \otimes_{\mathbb{F}_p} \bigotimes_{i=1}^n \mathcal{H}\mathcal{H}\left(\frac{\mathbb{F}_p[a_i]}{\mathbb{F}_p}\right)$$

is

$$\mathbb{F}_p[u] \otimes_{\mathbb{F}_p} \bigotimes_{i=1}^n \wedge_{\mathbb{F}_p[a_i]}(\sigma a_i) \rightarrow \mathbb{F}_p[u^{\pm 1}] \otimes_{\mathbb{F}_p} \bigotimes_{i=1}^n \wedge_{\mathbb{F}_p[a_i]}(\sigma a_i)$$

$$\mathbb{F}_p[u, a_1, \dots, a_n] \otimes \wedge(\sigma a_1, \dots, \sigma a_n) / (a_1, \dots, a_n)$$

$$\mathbb{F}_p[u, a_1, \dots, a_n] \otimes \Lambda(\sigma a_1, \dots, \sigma a_n) / (a_1, \dots, a_n) \\ \longrightarrow \mathbb{F}_p[u^{\pm 1}, a_1, \dots, a_n] \otimes \Lambda(\sigma a_1, \dots, \sigma a_n) / (a_1, \dots, a_n)$$

$$u^{\hat{j}} (\sigma a_1)^{\hat{\sigma}_1} \dots (\sigma a_n)^{\hat{\sigma}_n} \quad \text{where } j < 0.$$

$|a_1|+1$ $|a_n|+1$

$$* > \sum_{i=1}^n |a_i| + 1 = n + \sum_{i=1}^n |a_i|$$

such elements do not appear.

□

$$\mathbb{B}_p(n) [x]_{/x^e}, \quad \mathbb{B}_p(n) [x, y]_{/ (y^2 - x^3)}$$

$$\mathbb{B}_p(n) [M]$$

M monoid